

23/7/26

CO1

Assessment
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SHORT

ANSWERS

1)

Given :

$$\text{Total Spectrum} = 45 \text{ MHz} = 45,000 \text{ kHz}$$

$$\left. \begin{array}{l} \text{Simplex channel} \\ \text{Bandwidth} \end{array} \right\} = 25 \text{ kHz}$$

$$\left. \begin{array}{l} \text{One Full-Duplex} \\ \text{channel} \end{array} \right\} = 2 \text{ Simplex channels}$$

Formula :

$$\begin{aligned} * \text{ No. of Simplex Channels} &= \frac{\text{Total Spectrum}}{\text{channel Bandwidth}} \\ &= 45,000 / 25 = 1800 \end{aligned}$$

$$\begin{aligned} * \text{ No. of Full duplex channels} &= \frac{\text{No. of Simplex channels}}{2} \\ &= 1800 / 2 = 900 \end{aligned}$$

$\therefore$ The Spectrum can support 900 full duplex voice channels.

2)

$$\begin{aligned} \text{Given : } \text{Total voice channels} &= 560 \\ \text{cluster size} &= 7 \end{aligned}$$

$$\text{Formula : } \text{channels per cell} = \frac{\text{Total channels}}{\text{cluster size}}$$

$$\text{channels per cell} = 560 / 7 = 80$$

$$\therefore 80 \text{ voice channels.}$$

3,

$$\begin{aligned} \text{Given : } \text{No. of Carriers} &= 100 \\ \text{Time slots per Carrier} &= 8. \end{aligned}$$

$$\text{Formula : } \text{Total voice channels} \times \text{Time Slots}$$

$$100 \times 8 = 800$$

$\therefore$ The GSM deployment supports 800 simultaneous voice channels.

4,

Given: Base Station Power = 80W

Handset Power = 4W

Formula: Power Ratio (dB) = $10 \log_{10}(P_1/P_2)$

$$\text{calculation} = 10 \log_{10}(80/4)$$

$$= 10 \log_{10}(20)$$

$$= 10 \times 1.301$$

$$= 13.01 \text{ dB}$$

$\therefore$ The power control Ratio is 13.01dB.

5,

Given: Carrier Power = -60dBm

Interference Power = -92dBm

Formula: C/I = Carrier Power - Interference Power

$$= -60 - (-92)$$

$$= 32 \text{ dB}$$

6,

Given: Total Spectrum = 42MHz

Channel bandwidth = 300 KHz

Cluster Size = 7

Formula: total channels = $\frac{\text{Spectrum}}{\text{channel width}}$

$$= \frac{142,000}{300} = 140$$

$$\text{Channels per cell} = \frac{140}{\text{cluster size}} = \frac{140}{7} = 20$$

$\therefore$ Result: 20 channels per cell.

7,

Given Data :

Initial power = 64 W

Path-loss exponent = 4

Formula: $P_2 = P_1 \left(\frac{d_1}{d_2}\right)^4$

$$= 64 \cdot \left(\frac{1}{2}\right)^4$$

$$= 4 \text{ W.}$$

$\therefore$ The power becomes 4 W.

8,

Given Data :

Radius doubles

Inverse - Square law.

$$P \propto d^2 \Rightarrow (2)^2 = 4.$$

$\therefore$ Transmit Power must be increased by a factor of 4.

9,

Given Data :

Cluster Size = 4

Channels per cell = 15

Formula :

Total channels = Cells x

Channels per cell

$$= 4 \times 15 = 60.$$

$\therefore$ 60 channels.

10,

Given:

Speed = 100 km/h

Frequency = 900 MHz

$$f_d = \frac{v \cdot f}{c}$$

$$v = 27.78 \text{ m/s}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$f_d = \frac{27.7 \times 900 \times 10^6}{3 \times 10^8} = 83.3 \text{ Hz}$$

$$\therefore 83 \text{ Hz}$$

11,

Given Data:

$$\text{TDMA} = 5 \text{ ms}$$

$$\text{No. of time Slots} = 8$$

$$\text{Time Slot Duration} = \frac{\text{Frame Duration}}{\text{No. of Slots}} = \frac{5}{8} = 0.625 \text{ ms}$$

12,

Given Data:

$$\text{total bandwidth} = 24 \text{ MHz}$$

$$\text{Carrier Spacing} = 200 \text{ kHz}$$

$$24 \text{ MHz} = 24,000 \text{ kHz}$$

$$= \frac{24,000}{200}$$

$$= 120$$

$\therefore 120$ carrier frequencies.

13,

Given Data:

$$\text{Transmit power} = 8 \text{ W}$$

$$\text{Path loss} = 33 \text{ dB}$$

$$P_r = P_t - L \quad 8 \text{ W} = 39 \text{ dBm}$$

$$39 - 33 = 6 \text{ dBm}$$

14, Given Frequency = 1900 MHz

Distance = 2.5 km

$$FSPL = 32.44 + 20 \log f + 20 \log d$$

$$= 32.44 + 65.58 + 7.96$$

$$= 105.98 \text{ dB.}$$

$\therefore$ Free Space path $\approx 106 \text{ dB.}$

15, Given Number of cells = 6

Calls per cell = 250

$$\text{Cluster Capacity} = \text{cells} \times \text{calls per cell}$$

$$= 6 \times 250$$

$$= 1500$$

16,

Given:

Call request = 4000 calls/s

Average call

duration = 3 minutes.

$$\text{Active calls} = \text{Call Rate} \times \text{Call Duration.}$$

$$3 \times 60 = 180 \text{ s}$$

$$4000 \times 180 = 720000$$

$\therefore$ 720000 active calls.

17,

Given:

no. of channels = 150

Bandwidth = 25 KHz.

$$150 \times 25 = 3750 \text{ KHz.}$$

$$= 3.75 \text{ MHz.}$$

18,

Given Data:

Frequency bands = 30

Guard band = 40 KHz.

no. of gaps: $30 - 1 = 29$

Bandwidth wasted : $29 \times 40 = 1160 \text{ KHz}$
 $= 1.16 \text{ MHz}.$

19,

Given:

Vehicle Speed = 90 km/h

Cell radius = 1.5 km.

Distance travelled = 90 km.

$$2 \times 1.5 = 3 \text{ km}$$

$$\frac{90}{3} = 30.$$

$\therefore 30$ Handoffs.

20,

Given:

Regional centres = 8.

Base Stations = 40

Users = 250.

$$\Rightarrow 8 \times 40 \times 250 = 80,000$$

$\therefore 80,000$ users.

21,

Given:

Total area = 400 km^2

Cell radius = 400 m = 0.4 km

Approx cell area = R^2

$$R = 0.4 \text{ km}.$$

$$\text{Area} = (0.4)^2 = 0.16 \text{ km}^2.$$

$$\text{no. of cells} = \frac{400}{0.16} = 2500.$$

22,

Given:

$$\text{bandwidth} = 6 \text{ MHz}$$

$$\text{channel BW} = 30 \text{ KHz.}$$

$$6 \text{ MHz} = 6000 \text{ KHz}$$

$$= \frac{6000}{30}$$

$$= 200.$$

$$\therefore \text{no. channels} = 200.$$

23,

Given:

$$\text{Subscribers} = 1200$$

Calls per

$$\text{Subscriber} = 4 \text{ calls/hour}$$

Average Call

$$\text{Duration} = 2.5 \text{ min.}$$

$$A = N \times \frac{\lambda + H}{60}$$

$$= \frac{4 \times 2.5}{60} = 0.1667$$

$$\begin{aligned} \text{Total Traffic} &= 1200 \times 0.1667 \\ &= 200. \end{aligned}$$

24,

Given:

$$\text{bandwidth} = 40 \text{ MHz}$$

$$\text{Bw per user} = 400 \text{ KHz.}$$

$$= \frac{40000}{400} = 100.$$

Max. users

25,

Given:

$$\text{Base Stations} = 1200$$

$$\text{Users per BS} = 600$$

$$1200 \times 600 = 720000$$

$$\therefore \text{Total user Capacity} = 720000 \text{ users.}$$

26,

Given:

$$\text{Total Cells} = 200$$

$$\text{Cells paged per round} = 4.$$

$$\text{Paging Rounds} = \frac{\text{Total Cells}}{\text{Cells per Round}}$$

$$= \frac{200}{4}$$

$$\therefore \text{Rounds} = 80. \Rightarrow 50.$$

27,

Given:

$$\text{System capacity} = 100,000 \text{ calls.}$$

$$\text{Call attempts} = 40,000 \text{ calls/min.}$$

$$\text{Avg. duration} = 3 \text{ mins.}$$

$$\text{offered calls, } 40,000 \times 3 = 120,000$$

$$\text{blocked calls, } 120,000 - 100,000 = 20,000$$

$$\text{blocking \%} = \frac{20,000}{120,000} \times 100 = 16.67\%$$

28,

Given:

$$\text{Total Subscribers} = 600,000$$

$$\text{Roaming} = 15\%$$

$$\Rightarrow \frac{15}{100} \times 600,000 = 90,000$$

29, Given:

Maximum capacity = 12,000 calls/s
Current load = 9,000 calls/s

$$= \frac{9000}{12,000} \times 100 = 75\%$$

30, $\therefore$ The switching center is operating at 75% utilization.

Given:

Total Spectrum = 48 MHz.

Channel Spacing = 200 kHz.

Frequencies = 4.

$$\begin{aligned} \text{Total channels} &= \frac{48,000}{200} \\ &= 240. \end{aligned}$$

$$\begin{aligned} \text{Per cell} &= \frac{240}{4} = 60. \end{aligned}$$

31,

Given: Total Spectrum = 30 MHz.

Channel Spacing = 200 kHz.

Convert bandwidth,

$$\begin{aligned} 30 \text{ MHz} &= 30,000 \text{ kHz} \\ &= \frac{30,000}{200} \end{aligned}$$

$$\begin{aligned} &= 150. \\ &150 \text{ GSM channels.} \end{aligned}$$

32, Given: Total handoff attempts = 2500
 Success rate = 99%
 $2500 \times 0.99 = 2475$
 $\therefore 2475$ Successful handoffs.

33, Given: Transmit power = 12W
 Path loss = 45dB.

Formula: $P_r = P_t - L$

Convert Transmit Power: $12W = 40.8\text{dBm}$.

Received Power: $40.8 - 45 = -4.2\text{dBm}$.

34, Given: Cluster Size = 7
 Users per cell = 900
 Formula: Capacity = Cells $\times$ Users per cell
 $7 \times 900 = 6300$
 $\therefore 6300$ Users.

35, Given: Frame duration = 4.8ms
 Slots = 8.
 Formula: Slot Duration = $\frac{\text{Frame Duration}}{\text{slots}}$
 $= \frac{4.8}{8}$
 $= 0.6\text{ms}$.

36,

Given:

Mobile Speed, $v = 150 \text{ km/h}$.Frequency, $f = 1.8 \text{ GHz}$.Speed of light, $c = 3 \times 10^8 \text{ m/s}$.

$$f_d = \frac{vf}{c}$$

$$1.8 \text{ GHz} = 1.8 \times 10^9 \text{ Hz}$$

$$f_d = \frac{41.67 \times 1.8 \times 10^9}{3 \times 10^8}$$

$$f_d \approx 250 \text{ Hz}$$

37,

Given:

Capacity = 250,000 calls

Call rate = 70,000 calls/min

Avg call = 4 mins.

Offered calls,

=) Call Rate $\times$ call Duration.

$$70,000 \times 4 = 280,000$$

$$280,000 - 250,000 = 30,000$$

$$\frac{30,000}{280,000} \times 100 = 10.71\%$$

38,

Given:

No. of cells = 9.

Users per cell = 600.

$$\text{Capacity} = \text{cells} \times \text{users/cell}$$

$$= 9 \times 600$$

$$= 5400$$

39,

Given: Calls = 60,000 calls/s

Drop rate = 0.4%.

$$\text{Dropped Call} = \frac{0.4}{100} \times 60,000$$

$$= 0.004 \times 60,000$$

$$= 240 \text{ calls/s}$$

40,

Given:

Transmit power = 43 dBm.

Received power = -87 dBm.

$$\text{Path Loss} = P_t - P_r$$

$$= 43 - (-87)$$

$$\therefore \text{path} = 130 \text{ dB}$$

$$\text{Loss} = 130 \text{ dB}$$

41,

Given:

$$BS = 2.5 \text{ km}^2$$

No. of base stations = 150.

$$\text{Total Area} = 150 \times 2.5$$

$$= 375 \text{ km}^2$$

42,

Given:

Total Spectrum = 48 MHz

Channel = 200 kHz.

Reuse factor = 6.

$$48 \text{ MHz} = 48,000 \text{ kHz}$$

$$= \frac{48,000}{200}$$

$$\frac{240}{6} = 40$$

$$= 240$$

43,

Given:

Users per cell = 6000

Total Subscribers = 150,000.

$$= \frac{150,000}{6000} = 25.$$

$\therefore$ 25 paging cycles.

44,

Given: Call duration = 9 mins.

Handoff interval = 3 ins.

$$= \frac{9}{3}$$

$$= 3.$$

$\therefore$ No. of Handoffs = 3.

45,

Given:

Signal power = -85 dBm

Noise power = -115 dBm

$$SNR = S - N$$

$$= -85 - (-115)$$

$$= 30 \text{ dB.}$$

46,

Given:

No. of users = 250

Bandwidth = 150 kbps.

$$\text{Total Bandwidth} = 250 \times 150$$

$$= 37,500 \text{ kbps}$$

$\therefore$ Minimum Bandwidth = 37.5 Mbps.

47,

Given: Subscribers = 1500

calls per Subscriber = 4 calls/hour.

Avg. call duration = 2.5 mins.

$$A = N \times \frac{\lambda \times H}{60} = \frac{4 \times 2.5}{60} = 0.1667$$

$$\therefore 1500 \times 0.1667 = 250 \text{ Erlangs}$$

48,

Given:

No. of control channels = 250

Bandwidth per channel = 12 kHz.

$$250 \times 12 = 3000 \text{ kHz} \\ = 3 \text{ MHz.}$$

$\therefore$ The Control Channels reserve 3 MHz of bandwidth.

49,

Given:

Simultaneous connections = 1500

Data rate per connection = 600 kbps.

$$1500 \times 600 = 900000 \text{ kbps,} \\ = 900 \text{ Mbps.}$$

$$\therefore \text{Total Throughput} = 900 \text{ Mbps.}$$

50,

Given:

Radius = 4 km

Total Service Area = 350 km²

$$A = \pi R^2 = 50.24 \text{ km}^2$$

$$\therefore \text{No. of towers} = \frac{350}{50.24} = 6.97 \quad \text{Round up} = 7 //$$